

IL6R marks immune-myeloid inflammatory tumour microenvironments in colorectal cancer despite limited evidence for circulating-protein causality

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Abstract

Background: IL-6/IL6R signalling is implicated in inflammatory regulation and tumour-microenvironment biology, but whether circulating IL6R has a detectable causal relationship with colorectal cancer (CRC) risk remains uncertain. We integrated genetic epidemiology, tumour transcriptomics, clinical endpoint reconstruction, single-cell RNA sequencing and network-support analyses to clarify how IL6R should be interpreted in CRC.

Methods: Two-sample Mendelian randomization (MR) evaluated genetically proxied circulating IL6R protein levels and CRC risk across a primary FinnGen endpoint and two replication CRC GWAS endpoints. Colocalization was performed at the IL6R locus. TCGA-COAD/READ analyses assessed IL6R expression, reconstructed overall survival (OS) and progression-free interval (PFI), and immune-infiltration correlations. GSE146771 single-cell RNA-seq data were used to localize IL6-axis expression and CellChat-supported ligand-receptor communication. IL6R-centred co-expression, enrichment and drug-target support analyses were used to contextualize immune, myeloid and inflammatory programmes.

Results: MR did not support a significant causal effect of genetically proxied circulating IL6R protein levels on CRC risk. Colocalization at the IL6R locus also did not support a strong shared causal variant (PP.H4 = 0.048). TCGA analyses showed markedly lower IL6R expression in tumours than adjacent normal tissues but did not support IL6R as a robust OS/PFI prognostic marker after patient-level

25 survival reconstruction. Manual external GEO validation expanded the independent validation layer to 1,107 samples across GSE14333, GSE17536, GSE17537 and GSE39582. In GSE39582, IL6R expression was significantly higher in non-tumoral colorectal mucosa than in tumour samples (median 6.677 vs 4.064; Wilcoxon $P = 3.94 \times 10^{-13}$). Stage-stratified analyses across four GEO cohorts and univariable survival models in GSE17536, GSE17537 and GSE39582 did not support IL6R as a
30 consistent stage-associated or standalone prognostic marker. Immune-infiltration, single-cell and support-layer analyses consistently linked IL6R to immune-, myeloid- and inflammatory-associated tumour microenvironment features.

Conclusions: Genetic analyses did not support a strong causal or colocalized effect of circulating IL6R protein levels on CRC risk, and TCGA plus external GEO survival checks did not support IL6R as a
35 robust standalone prognostic marker. The expanded GSE39582 tumour-normal validation further suggested that IL6R is not a tumour-intrinsic upregulated oncogene in CRC. Nevertheless, transcriptomic, single-cell and support-layer analyses consistently connected IL6R to an immune-, myeloid- and inflammatory-associated CRC tumour microenvironment.

Keywords: IL6R; interleukin-6; colorectal cancer; Mendelian randomization; colocalization; TCGA;
40 single-cell RNA-seq; tumour microenvironment; myeloid inflammation; drug-target network

Background

Colorectal cancer (CRC) remains a major global malignancy and a leading contributor to cancer morbidity and mortality. Recent GLOBOCAN estimates indicate that CRC accounts for more than 1.9 million incident cases and approximately 904,000 deaths worldwide, underscoring the need to clarify
45 molecular and immune-microenvironmental determinants of CRC biology [1]. Contemporary epidemiological updates also highlight persistent challenges related to late detection, recurrence, therapy resistance and early-onset disease [2]. These challenges make it important to distinguish

molecular features that are plausible causal determinants of CRC risk from features that instead reflect tumour ecology, immune infiltration or tissue-context responses.

50 Inflammatory signalling is central to CRC biology because chronic intestinal inflammation, stromal remodelling and immune-cell recruitment can shape epithelial transformation and tumour progression. Interleukin-6 (IL-6) signals through IL6R and the signal-transducing co-receptor IL6ST/gp130, activating downstream JAK/STAT3 and related pathways involved in immune regulation, survival signalling and inflammatory feedback [3,4]. However, IL-6-axis biology is context-dependent:

55 circulating cytokine or receptor levels, soluble receptor dynamics, membrane receptor expression and local cell-cell communication may represent different biological layers [5]. This creates an important interpretive challenge for IL6R in CRC.

Mendelian randomization (MR) can help evaluate whether genetically proxied molecular exposures are associated with disease risk under instrumental-variable assumptions [6,7]. Colocalization adds a

60 complementary genetic layer by evaluating whether exposure and outcome signals in a locus are compatible with a shared causal variant rather than neighbouring signals in linkage disequilibrium [9]. Together, MR and colocalization can separate putative genetic-causality evidence from broader tissue-level or tumour-microenvironment associations.

Negative MR or colocalization evidence does not imply that a molecule is biologically irrelevant.

65 Instead, it may indicate that the measured circulating exposure does not have a detectable germline causal effect on disease risk, whereas tissue-level expression or cell-specific signalling may still be associated with tumour states. TCGA-COAD/READ provides transcriptomic and clinical resources for tumour-normal expression, survival and immune-infiltration analyses [10-14]. Single-cell RNA sequencing further localizes expression and inferred ligand-receptor communication to tumour

70 microenvironment compartments [15-17]. Finally, co-expression, pathway-enrichment and drug-target

support layers can help determine whether IL6R-associated genes converge on immune, cytokine and myeloid programmes [18-21].

This study applied an integrated evidence-chain framework to IL6R in CRC. We first used MR and colocalization to test whether circulating IL6R protein levels were genetically associated with CRC risk and whether the IL6R locus supported shared causal variation. We then assessed IL6R expression, reconstructed OS/PFI endpoints and immune infiltration in TCGA-COAD/READ. Next, we used GSE146771 single-cell RNA-seq data to localize IL6-axis expression and communication. Finally, we constructed IL6R-centred co-expression, enrichment and drug-target support layers. The central aim was not to force a causal interpretation where genetic evidence is weak, but to position IL6R within an explicit evidence hierarchy: genetic-causality testing first, followed by transcriptomic and cellular-context interpretation.

Methods

Overall study design

This study was designed as an integrative multi-layer analysis to evaluate the relationship between IL6R-related biology and CRC. The overall workflow and evidence-chain logic are summarized in Figure 1. The analytical framework included two-sample MR, IL6R-locus colocalization, TCGA-COAD/READ transcriptomic and clinical analyses, GSE146771 single-cell RNA-seq analysis and support-layer network, enrichment and drug-target annotation. Because the MR and colocalization layers did not support a strong causal or shared-variant effect, downstream transcriptomic and network analyses were interpreted as biological-context support rather than as proof of causal therapeutic targeting.

Data sources and analytical layers were organized as follows. Genetic exposure analyses used circulating IL6R protein pQTL instruments to test genetic causal evidence. CRC outcome analyses used FinnGen CRC plus IEU and EBI/GWAS Catalog CRC endpoints for primary and replication MR

95 analyses. Colocalization used IL6R-locus exposure and outcome regional statistics to evaluate shared
causal-variant support. TCGA transcriptomic analyses used TCGA-COAD/READ expression and
clinical data to evaluate tumour expression, reconstructed OS/PFI and immune context. Single-cell
analysis used GSE146771 10x and Smart-seq2 data to localize cell-state and IL6-axis communication
evidence. The support layer used co-expression, GO/KEGG/Reactome and drug-target annotation to
100 contextualize immune/myeloid programmes.

Genetic exposure and outcome data for Mendelian randomization

Genetic instruments were prepared for IL6R and related inflammatory traits, including IL-6 and CRP
where available, using harmonized summary-level genetic association files preserved in the project
workflow. The primary exposure for manuscript interpretation was genetically proxied circulating
105 IL6R protein level. CRC outcome analyses included a FinnGen colorectal cancer endpoint and two
additional CRC-related GWAS endpoints used for replication and robustness checking. Harmonized
output files included IL6R-to-FinnGen CRC, IL6R-to-IEU CRC and IL6R-to-EBI CRC analyses,
allowing the primary and replication endpoints to be interpreted on a comparable analysis scale.

Genetic variants were retained as instrumental variables only when they satisfied the pre-specified
110 relevance and harmonization criteria in the MR workflow. Palindromic and ambiguous variants were
handled during harmonization, and alleles were aligned so that causal estimates corresponded to the
same effect allele for exposure and outcome. The MR workflow followed the conceptual basis of
instrumental-variable causal inference in genetic epidemiology and used multiple sensitivity analyses
to reduce over-reliance on a single estimator [6,7].

115 Mendelian randomization and sensitivity analyses

The primary MR estimator was inverse-variance weighting (IVW). Weighted median, MR-Egger and
weighted mode analyses were used as complementary estimators. Heterogeneity was assessed using
Cochran's Q statistics, and horizontal pleiotropy was assessed using MR-Egger intercept tests. Leave-

one-out analyses were used to evaluate whether the primary estimate was driven by a single variant.

120 Results were reported as beta coefficients, odds ratios (ORs), 95% confidence intervals (CIs), P values and number of SNPs. MR reporting was aligned with STROBE-MR principles, emphasizing transparent reporting of data sources, assumptions, sensitivity analyses and limitations [8].

Colocalization at the IL6R locus

Colocalization was performed at the IL6R locus using exposure-region and outcome-region summary
125 statistics preserved in the project release. The analysis used an approximate Bayesian framework to estimate posterior probabilities for five hypotheses: no association with either trait, association with the exposure only, association with the outcome only, association with both traits through different causal variants, and association with both traits through a shared causal variant. The key interpretive statistic was PP.H4, the posterior probability for a shared causal variant [9]. PP.H4 was interpreted
130 conservatively; low PP.H4 was treated as evidence against strong shared-variant support rather than as proof that IL6R has no biological role in CRC.

TCGA-COAD/READ expression and clinical analyses

TCGA-COAD/READ transcriptomic and clinical analyses were performed using project-preserved
TCGA data objects and output tables. IL6R expression was compared between adjacent normal and
135 tumour tissues in combined COAD/READ and in COAD and READ strata. The project workflow used TCGA data structures generated through GDC/TCGA-compatible preprocessing, with TCGAbiolinks-style TCGA integration serving as the reference computational framework [10,11].

For clinical analysis, patient-level OS and PFI endpoints were reconstructed after an earlier survival-
summary issue was identified. IL6R expression was matched to patient barcodes, and PFI information
140 was supplemented from the TCGA Pan-Cancer Clinical Data Resource where required [12].

Continuous Cox models were used as the primary survival association analysis, while median-split
Kaplan-Meier curves were generated as visual summaries. OS and PFI were assessed in the combined

COAD/READ cohort and in COAD and READ strata. Results were interpreted as clinical association evidence, not as causal evidence.

145 Immune-infiltration analyses in TCGA

To contextualize IL6R expression in relation to tumour microenvironment composition, immune and stromal infiltration features were evaluated using MCP-counter and quanTIseq-style immune deconvolution outputs [13,14]. Spearman correlations were calculated between IL6R expression and immune-infiltration scores in COAD and READ. Immune-infiltration results were interpreted with
150 emphasis on concordant patterns across myeloid, neutrophil, macrophage/monocyte, endothelial and T-cell-related signatures rather than single isolated cell types.

Single-cell RNA-seq and cell-cell communication analyses

Single-cell RNA-seq analyses were performed using the GSE146771 10x and Smart-seq2 datasets preserved in the project workflow. The analysis included quality-controlled expression matrices,
155 dimensionality reduction, cell-state visualization, IL6-axis dot plots, module score calculation and cell-type-level summaries. Seurat-style single-cell workflows were used as the conceptual framework for normalization, clustering and visualization [15,16].

CellChat was used to infer ligand-receptor communication networks from single-cell expression data, with specific attention to IL6 -> IL6R_IL6ST signalling [17]. Smart-seq2 results were treated as the
160 primary IL6-axis communication evidence because the Smart-seq2 workflow recovered IL6-axis communication edges, whereas the 10x workflow did not recover IL6-axis edges despite successful CellChat execution. The 10x result was interpreted as a sparse-platform comparison rather than proof of absent IL6-axis communication.

IL6R-centred co-expression, enrichment and drug-target support analyses

165 A support-layer analysis was performed to contextualize IL6R-associated transcriptomic biology in TCGA tumour samples. Spearman correlations between IL6R expression and genome-wide tumour expression were calculated, and an IL6R-associated support gene set was defined using positively correlated genes and manually curated immune, myeloid and inflammatory seed genes. A co-expression network was generated from the support gene set. Functional enrichment was performed
170 using Gene Ontology Biological Process, KEGG and Reactome pathway resources through clusterProfiler-compatible workflows [18-20,34]. STRING and related network concepts were used only as background support for interpreting protein and pathway networks, not as direct evidence of causal biology [21].

Drug-target annotation was used to summarize candidate axis-level relationships involving
175 IL6/IL6R/JAK/STAT and myeloid-associated targets. These relationships were interpreted as exploratory druggability context only. They were not interpreted as evidence that the listed agents are clinically validated CRC therapies or that IL6R blockade would necessarily benefit CRC patients.

Statistical analysis and reproducibility

All analyses were conducted using scripted R workflows and versioned output folders. Statistical tests
180 were two-sided unless otherwise specified. Multiple-testing correction was applied where implemented in the workflow, including false discovery rate adjustment for tumour-normal expression and enrichment analyses. Processed supplementary tables, figure-source manifests, GEO validation scripts and workflow records are archived in the GitHub release listed in the Availability of data and materials section.

185 External GEO validation

External GEO validation was added as an independent validation layer to test whether IL6R expression showed reproducible tumour-normal, stage-stratified or prognostic behaviour across external CRC

microarray cohorts. GSE14333, GSE17536, GSE17537 and GSE39582 were queried and curated using GEO series-matrix metadata, GPL570 probe annotations and project-preserved validation outputs.

190 IL6R probes were identified from GPL570 annotations, and the primary IL6R probe was used for sample-level expression summaries. Manual phenotype curation was then used to map tissue class, AJCC/TNM or Dukes stage, and available OS, DFS/RFS or DSS endpoints. Tumour-normal validation was performed only in cohorts with explicit sample-level tumour and non-tumour labels. Stage validation compared early-stage (I-II) and advanced-stage (III-IV) samples. Univariable Cox

195 proportional-hazards models used z-scored IL6R expression and were interpreted as external robustness checks rather than as definitive clinical prognostic models.

Results

Mendelian randomization did not support a significant causal effect of circulating IL6R on CRC risk

Two-sample MR was performed to evaluate whether genetically proxied circulating IL6R protein

200 levels were associated with CRC risk. In the primary FinnGen CRC endpoint, the IVW estimate did not support a statistically significant association (beta = -0.0474, OR = 0.954, 95% CI 0.889-1.024, P = 0.190). Complementary weighted median, MR-Egger and weighted mode analyses were directionally and statistically consistent with a null or weak association rather than a robust causal effect.

Replication endpoints also did not provide significant support: the IEU CRC endpoint yielded IVW P

205 = 0.916, and the EBI Firth-corrected CRC endpoint yielded IVW P = 0.603. The MR summary with readable endpoint labels is shown in Figure 2.

Sensitivity analyses did not reveal strong evidence of violation of MR assumptions in the primary endpoint. In FinnGen CRC, heterogeneity and MR-Egger intercept tests did not indicate strong heterogeneity or directional pleiotropy, and leave-one-out analyses did not identify a single SNP that

210 converted the overall estimate into a robust significant association. These findings indicate that the

available genetic evidence did not support a strong causal effect of circulating IL6R protein levels on CRC risk.

Colocalization did not support a shared causal variant at the IL6R locus

Colocalization analysis was used to evaluate whether the IL6R protein-associated signal and the CRC
215 risk-associated signal at the IL6R locus were compatible with a shared causal variant. The analysis
included 1,062 SNPs and yielded $PP.H4 = 0.048$. This value was well below the threshold typically
interpreted as strong support for colocalization. The posterior probability pattern was instead
dominated by exposure-only evidence, indicating that the IL6R pQTL signal and the CRC association
signal were not strongly supported as arising from the same causal variant. Therefore, the genetic
220 evidence did not support a shared-variant mechanism linking circulating IL6R protein levels to CRC
risk (Figure 3).

IL6R expression was markedly reduced in CRC tumours but was not a robust OS/PFI prognostic
marker

In TCGA-COAD/READ, IL6R expression was significantly lower in tumour tissues than in adjacent
225 normal tissues. In the combined cohort, median IL6R expression was 11.752 in normal tissues and
8.353 in tumours, with $FDR = 1.09e-31$. Similar tumour-normal decreases were observed in COAD
and READ strata, supporting a consistent reduction of IL6R mRNA expression in CRC tumour tissue.

Because an earlier survival summary produced missing survival estimates, OS and PFI were
reconstructed at the patient level. After this repair, IL6R expression was not significantly associated
230 with OS or PFI. In the combined COAD/READ cohort, OS analysis included 551 patients and 121
events (log-rank $P = 0.968$; continuous Cox HR = 0.931, 95% CI 0.754-1.151, $P = 0.511$), while PFI
analysis included 596 patients and 153 events (log-rank $P = 0.546$; continuous Cox HR = 1.052, 95%
CI 0.871-1.271, $P = 0.599$). Similar null associations were observed in COAD and READ strata. These

results indicate that IL6R is not a robust standalone prognostic marker in TCGA CRC, despite its
235 strong tumour-normal expression difference (Figure 4).

IL6R expression was associated with immune-infiltrated and myeloid-enriched tumour
microenvironments

IL6R expression showed strong associations with immune-infiltration features in TCGA CRC. MCP-
counter analyses showed particularly strong correlations between IL6R expression and endothelial,
240 neutrophil, macrophage/monocyte, myeloid dendritic-cell and T-cell-related signatures. In COAD,
representative Spearman correlations included endothelial cell $\rho = 0.679$, neutrophil $\rho = 0.676$,
macrophage/monocyte $\rho = 0.668$ and T-cell $\rho = 0.653$. In READ, representative correlations
included neutrophil $\rho = 0.719$, endothelial cell $\rho = 0.717$ and macrophage/monocyte $\rho = 0.685$.
quanTIseq results provided additional immune-context support. These results suggest that IL6R
245 expression in bulk CRC transcriptomes reflects immune and stromal composition, particularly myeloid
and inflammatory tumour microenvironment features (Figure 5).

Single-cell analysis localized IL6-axis activity in the CRC tumour microenvironment

GSE146771 single-cell RNA-seq analysis included both 10x and Smart-seq2 data. The 10x dataset
included 14,662 cells and 13,538 features, and the Smart-seq2 dataset included 10,322 cells and 15,179
250 features. UMAP visualizations, IL6-axis module-score summaries and cell-communication outputs
were generated for both platforms.

CellChat analysis supported platform-dependent IL6-axis communication. In Smart-seq2, CellChat
detected 35,283 total communication edges and 32 IL6-axis edges, including IL6 -> IL6R_IL6ST
interactions. In contrast, the 10x analysis successfully ran CellChat but did not recover IL6-axis edges.
255 The Smart-seq2 results were therefore treated as the primary evidence for IL6-axis communication,
while the 10x results were interpreted as a sparse-platform comparison. Together, these findings

support the presence of IL6-axis activity in specific CRC tumour microenvironment contexts (Figure 6).

IL6R-centred support-layer analyses reinforced immune, myeloid and inflammatory biology

260 To further characterize the transcriptomic context of IL6R in CRC, an IL6R-centred tumour co-expression network was constructed from TCGA-COAD/READ tumour samples. The resulting support-layer network contained 109 support genes, 109 nodes and 4,933 co-expression edges. Functional enrichment analysis identified 895 Gene Ontology Biological Process records, 54 KEGG records and 227 Reactome records. The strongest terms converged on immune-response activation, 265 leukocyte activation, chemokine and cytokine signalling, myeloid-cell activation and inflammatory response programmes.

Exploratory drug-target annotation was used to map IL6/IL6R/JAK/STAT and myeloid-associated nodes to candidate drug-target relationships. Anti-IL6R agents, anti-IL6 therapy, JAK inhibitors, a STAT3-pathway inhibitor and a CSF1R inhibitor were represented as exploratory axis-level 270 candidates. These results should be interpreted as hypothesis-generating support for IL6R-associated immune/myeloid biology, rather than as evidence of clinically validated CRC therapy (Figure 7).

Manual external GEO validation supported a non-tumour-intrinsic and non-prognostic interpretation of IL6R expression

Manual GEO phenotype validation included 1,107 samples across four CRC microarray cohorts:

275 GSE14333 (n = 290), GSE17536 (n = 177), GSE17537 (n = 55) and GSE39582 (n = 585). In GSE39582, which included 566 tumour samples and 19 non-tumoral colorectal mucosa samples, IL6R expression was significantly higher in non-tumoral mucosa than in tumours (median 6.677 vs 4.064; median difference tumour minus normal = -2.613; Wilcoxon P = 3.94×10^{-13}). Stage-stratified analyses did not show significant IL6R differences between early-stage and advanced-stage tumours in

280 GSE14333 (P = 0.635), GSE17536 (P = 0.639), GSE17537 (P = 0.261) or GSE39582 (P = 0.464).
Univariable Cox models also did not support IL6R as a robust standalone prognostic marker:
GSE17536 OS HR = 0.897 (95% CI 0.709-1.135, P = 0.366), GSE17537 OS HR = 1.323 (95% CI
0.865-2.024, P = 0.196), GSE39582 OS HR = 0.993 (95% CI 0.867-1.138, P = 0.922), GSE39582
DFS/RFS HR = 0.928 (95% CI 0.786-1.096, P = 0.380), and the remaining available DFS/DSS models
285 were likewise non-significant. These external results strengthened the conservative interpretation that
IL6R is not a consistent tumour-stage or survival biomarker, while supporting a tissue-context model
in which IL6R is more abundant in non-tumoral inflammatory mucosal context than in tumour tissue.

Discussion

This integrative study separated genetic-causality evidence from transcriptomic and single-cell
290 evidence to clarify the role of IL6R in CRC. The main finding is deliberately conservative: genetically
proxied circulating IL6R protein levels were not significantly associated with CRC risk in MR
analyses, and colocalization did not support a shared causal variant at the IL6R locus. These results
argue against a strong germline causal interpretation for circulating IL6R in CRC risk under the
available instruments and outcome definitions. At the same time, TCGA and single-cell analyses
295 consistently linked IL6R to immune-infiltrated, myeloid-enriched and inflammatory CRC tumour
microenvironments. This distinction is central to the manuscript: IL6R should not be framed as a
proven causal CRC-risk driver, but as an immune-context marker and potential signalling node within
tumour-microenvironment biology.

The negative MR and colocalization results are biologically informative. MR is useful for testing
300 whether lifelong genetically proxied differences in an exposure influence disease risk, but it does not
capture every biological context of a molecule [6,7]. In this study, the exposure represented circulating
IL6R protein levels, whereas tumour biology may depend on membrane-bound receptor expression,
soluble receptor dynamics, local IL-6 production, IL6ST availability, stromal interactions and cell-

type-specific signalling. Colocalization further indicated that the IL6R pQTL signal and CRC risk
305 signal were not strongly supported as the same causal variant, making a direct shared-variant claim
inappropriate [9]. This combination of results supports a cautious interpretation: the available genetic
evidence does not establish circulating IL6R as a causal determinant of CRC risk.

The TCGA tumour-normal result provides a different layer of evidence. IL6R mRNA expression was
markedly lower in CRC tumours than in adjacent normal tissues, yet this expression difference did not
310 translate into a robust OS/PFI prognostic association after patient-level survival reconstruction. This
pattern suggests that IL6R expression may reflect tissue composition or immune/stromal architecture
rather than tumour-cell-intrinsic aggressiveness. The use of TCGA-CDR-supported PFI reconstruction
was important because clinical endpoint quality strongly affects prognostic interpretation in TCGA
analyses [12]. The repaired survival findings therefore refine the claim: IL6R is not a strong standalone
315 survival marker in this cohort, even though it remains biologically meaningful in tumour immune-
context analysis.

The immune-infiltration results provide the strongest tissue-level explanation for this pattern. IL6R
expression correlated with endothelial, neutrophil, macrophage/monocyte, myeloid dendritic-cell and
T-cell-related signatures, pointing to an immune-infiltrated and myeloid-enriched tumour
320 microenvironment. This is consistent with the broader biology of the IL-6/IL6R/JAK/STAT3 axis,
which can shape cytokine signalling, immune-cell recruitment, stromal activation, angiogenesis and
inflammatory feedback [3,4,29,30]. It also explains why IL6R may be transcriptomically informative
despite negative MR, colocalization and survival-marker results: bulk IL6R expression may partly
represent immune and stromal cellular composition rather than an isolated tumour-cell oncogenic
325 dependency.

Single-cell results strengthened the cellular-context interpretation. GSE146771 analysis showed that
IL6-axis expression and module scores could be localized across tumour microenvironment

compartments. Smart-seq2 CellChat analysis detected IL6 -> IL6R_IL6ST interactions, supporting ligand-receptor communication through the canonical IL6R/IL6ST receptor complex [17]. The absence
330 of recovered IL6-axis edges in 10x data should be interpreted carefully because single-cell platforms differ in sparsity, transcript capture and sensitivity [15,16]. For this reason, Smart-seq2 was used as the primary CellChat communication evidence, while 10x was treated as a conservative platform comparison.

The support-layer network, enrichment and drug-target analyses further contextualized IL6R-
335 associated biology. The IL6R-centred co-expression network and GO/KEGG/Reactome enrichment outputs converged on immune activation, cytokine-mediated signalling, myeloid leukocyte activation, chemotaxis and inflammatory pathways. These results align with the TCGA immune-infiltration and single-cell communication findings and support a coherent tumour-microenvironment model [18-21]. The drug-target network placed IL6R within an IL6/IL6R/JAK/STAT and myeloid-associated axis.
340 However, these annotations are exploratory. They are best viewed as druggability context for an immune/myeloid axis, not as evidence that the listed agents are established CRC therapies or that targeting IL6R would necessarily improve CRC outcomes.

The external GEO validation layer reduced one important weakness of the original computational study by adding independent CRC cohorts outside TCGA and, after manual phenotype curation,
345 extending beyond survival-only validation. The GSE39582 tumour-normal result externally supported the TCGA observation that IL6R is not tumour-upregulated in CRC; indeed, IL6R expression was substantially higher in non-tumoral colorectal mucosa than in tumour samples. Across four GEO cohorts, stage-stratified analyses did not show consistent early-versus-advanced stage differences, and univariable OS/DFS/DSS models did not support IL6R as a standalone prognostic marker. This pattern
350 is coherent with the manuscript's central model: IL6R is best interpreted as an immune/inflammatory microenvironmental feature rather than a direct tumour-intrinsic causal or prognostic driver.

Several limitations should be acknowledged. First, MR estimates depend on the strength, specificity and biological interpretation of available IL6R protein instruments. Circulating IL6R may not fully capture tissue-specific receptor biology, soluble versus membrane-bound receptor effects, or local IL6R/IL6ST signalling. Second, colocalization depends on locus coverage, ancestry compatibility, phenotype definitions and the single-causal-variant assumptions of the approximate Bayes factor framework [9]. Third, TCGA analyses are observational and cannot establish causal direction. Bulk immune-infiltration scores are computational estimates and may be influenced by tumour purity, stromal content and compositional effects. Fourth, single-cell analyses are affected by sample composition, platform sparsity, cell-state annotation and ligand-receptor inference assumptions. Fifth, the drug-target analysis is curated and exploratory rather than pharmacological validation. Finally, the present study requires public deposition of processed outputs and analysis code before final submission to maximize reproducibility.

The independent GEO validation also has limitations. It was based on public microarray cohorts, probe-level IL6R measurements, manually curated phenotype fields and univariable Cox models. GSE39582 contributed the only reliable tumour-normal external comparison, and the normal mucosa group was small (n = 19). Stage mappings relied on public AJCC/TNM or Dukes annotations, and RFS fields were treated as DFS-like endpoints where appropriate. These results should therefore be interpreted as external robustness checks, not as definitive clinical validation. Future work should test IL6R-axis activity in well-annotated prospective CRC cohorts, ideally with paired tumour-normal profiling, spatial or protein-level validation, multivariable clinical modelling, and treatment-response endpoints.

Overall, the findings support a revised model of IL6R biology in CRC. IL6R is not supported by the present genetic evidence as a causal CRC risk factor, nor by repaired TCGA survival analyses as a standalone prognostic marker. Instead, IL6R appears to mark or participate in immune-infiltrated, myeloid-enriched and inflammatory CRC tumour microenvironments. Future studies should test

whether IL6R-associated immune states predict immunotherapy response, inflammatory molecular subtypes, spatial immune organization, stromal niches or treatment-specific vulnerabilities in independent CRC cohorts and experimental systems.

380 Conclusions

This integrative analysis did not support a strong genetic causal or colocalized effect of circulating IL6R protein levels on CRC risk. Repaired TCGA survival analyses and external GEO survival checks also did not support IL6R as a robust standalone prognostic marker. However, TCGA immune-infiltration, single-cell and support-layer analyses consistently linked IL6R to an immune-, myeloid-
385 and inflammatory-associated CRC tumour microenvironment. IL6R should therefore be interpreted primarily as a context-dependent microenvironmental marker and pathway component rather than as a genetically supported causal risk factor or isolated prognostic biomarker in CRC.

Abbreviations

CRC: colorectal cancer; MR: Mendelian randomization; IVW: inverse-variance weighted; OR: odds
390 ratio; CI: confidence interval; TCGA: The Cancer Genome Atlas; COAD: colon adenocarcinoma; READ: rectum adenocarcinoma; OS: overall survival; PFI: progression-free interval; scRNA-seq: single-cell RNA sequencing; TME: tumour microenvironment; GO: Gene Ontology; KEGG: Kyoto Encyclopedia of Genes and Genomes; PP.H4: posterior probability of a shared causal variant.

Declarations

395 Ethics approval and consent to participate

This study used publicly available, de-identified summary-level GWAS data, TCGA-derived bulk transcriptomic and clinical data, TCGA-CDR survival annotations and public single-cell RNA-seq data. No new human participants, biospecimens or animal experiments were involved. Additional

ethics approval and consent to participate were therefore not required for this secondary analysis of
400 public resources.

Consent for publication

Not applicable.

Availability of data and materials

All analyses used publicly available or project-processed derivatives of public resources. GWAS
405 summary statistics were obtained from FinnGen, IEU/OpenGWAS and EBI/GWAS Catalog sources as
specified in the Methods. TCGA analyses used publicly available TCGA-COAD/READ and TCGA-
CDR resources. Single-cell analyses used the public GSE146771 dataset. External validation used
public GEO datasets GSE14333, GSE17536, GSE17537 and GSE39582. Project-processed tables,
figure source data, GEO validation outputs and workflow records are available in the GitHub release
410 supporting this manuscript:
https://github.com/1209433622cz-maker/MR_IL6R_CRC_project/releases/tag/v1.0.0. Public datasets
analyzed in this study are available from their original repositories as cited in the manuscript.

Competing interests

The author declares no competing interests.

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Authors' contributions

ZC conceived and designed the study, performed the computational analyses, interpreted the results,
prepared the figures and tables, drafted the manuscript and approved the submitted version.

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Additional file 4: GEO validation scripts, Figure 6 rerun script and workflow records. File format: R, PowerShell and Markdown.

Additional file 3: Figure source-data and traceability registries, including Supplementary Table S8 linking each figure to detected source-data files and code paths. File format: CSV/TSV/ZIP archives.

Additional file 2: Supplementary Tables S1-S12. GEO validation summary, probe audit, sample-level expression, Cox validation models, phenotype-column audit, tumour-normal validation, figure source-data/code crosswalk and manual GEO validation tables. File format: TSV/CSV.

Additional file 1: Supplementary Figures S1-S2. External GEO validation of IL6R expression, tumour-normal comparison, stage-stratified validation and survival models. File format: PNG/PDF/SVG.

Additional files

Artificial intelligence-assisted language editing and formatting support was used during manuscript development and workflow documentation. The author reviewed and verified all analyses, results, interpretations and final manuscript content, and takes full responsibility for the work.

Use of artificial intelligence tools

References

Bray F, Laversanne M, Sung H, Ferlay J, Siegel RL, Soerjomataram I, et al. Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA Cancer J Clin. 2024;74(3):229-263. doi:10.3322/caac.21834.

Siegel RL, Wagle NS, Cercek A, Smith RA, Jemal A. Colorectal cancer statistics, 2023. CA Cancer J Clin. 2023;73(3):233-254. doi:10.3322/caac.21772.

- Waldner MJ, Foersch S, Neurath MF. Interleukin-6 - a key regulator of colorectal cancer development. Int J Biol Sci. 2012;8(9):1248-1253. doi:10.7150/ijbs.4614.
- Johnson DE, O’Keefe RA, Grandis JR. Targeting the IL-6/JAK/STAT3 signalling axis in cancer. Nat Rev Clin Oncol. 2018;15(4):234-248. doi:10.1038/nrclinonc.2018.8.
- Hirano T. IL-6 in inflammation, autoimmunity and cancer. Int Immunol. 2021;33(3):127-148. doi:10.1093/intimm/dxaa078.
- Davey Smith G, Ebrahim S. “Mendelian randomization”: can genetic epidemiology contribute to understanding environmental determinants of disease? Int J Epidemiol. 2003;32(1):1-22. doi:10.1093/ije/dyg070.
- Hemani G, Zheng J, Elsworth B, Wade KH, Haberland V, Baird D, et al. The MR-Base platform supports systematic causal inference across the human phenome. eLife. 2018;7:e34408. doi:10.7554/eLife.34408.
- Skrivankova VW, Richmond RC, Woolf BAR, Davies NM, Swanson SA, VanderWeele TJ, et al. Strengthening the Reporting of Observational Studies in Epidemiology Using Mendelian Randomization: the STROBE-MR statement. JAMA. 2021;326(16):1614-1621. doi:10.1001/jama.2021.18236.
- Giambartolomei C, Vukcevic D, Schadt EE, Franke L, Hingorani AD, Wallace C, et al. Bayesian test for colocalisation between pairs of genetic association studies using summary statistics. PLoS Genet. 2014;10(5):e1004383. doi:10.1371/journal.pgen.1004383.
- Cancer Genome Atlas Network. Comprehensive molecular characterization of human colon and rectal cancer. Nature. 2012;487(7407):330-337. doi:10.1038/nature11252.

- 465 Colaprico A, Silva TC, Olsen C, Garofano L, Cava C, Garolini D, et al. TCGAAbiolinks: an R/Bioconductor package for integrative analysis of TCGA data. *Nucleic Acids Res.* 2016;44(8):e71. doi:10.1093/nar/gkv1507.
- Liu J, Lichtenberg T, Hoadley KA, Poisson LM, Lazar AJ, Cherniack AD, et al. An integrated TCGA pan-cancer clinical data resource to drive high-quality survival outcome analytics. *Cell.* 2018;173(2):400-416.e11. doi:10.1016/j.cell.2018.02.052.
- 470 Becht E, Giraldo NA, Lacroix L, Buttard B, Elarouci N, Petitprez F, et al. Estimating the population abundance of tissue-infiltrating immune and stromal cell populations using gene expression. *Genome Biol.* 2016;17(1):218. doi:10.1186/s13059-016-1070-5.
- Finotello F, Mayer C, Plattner C, Laschober G, Rieder D, Hackl H, et al. Molecular and pharmacological modulators of the tumor immune contexture revealed by deconvolution of RNA-seq data. *Genome Med.* 2019;11(1):34. doi:10.1186/s13073-019-0638-6.
- 475 Butler A, Hoffman P, Smibert P, Papalexi E, Satija R. Integrating single-cell transcriptomic data across different conditions, technologies, and species. *Nat Biotechnol.* 2018;36(5):411-420. doi:10.1038/nbt.4096.
- 480 Stuart T, Butler A, Hoffman P, Hafemeister C, Papalexi E, Mauck WM 3rd, et al. Comprehensive integration of single-cell data. *Cell.* 2019;177(7):1888-1902.e21. doi:10.1016/j.cell.2019.05.031.
- Jin S, Guerrero-Juarez CF, Zhang L, Chang I, Ramos R, Kuan CH, et al. Inference and analysis of cell-cell communication using CellChat. *Nat Commun.* 2021;12(1):1088. doi:10.1038/s41467-021-21246-9.
- 485 Wu T, Hu E, Xu S, Chen M, Guo P, Dai Z, et al. clusterProfiler 4.0: a universal enrichment tool for interpreting omics data. *Innovation (Camb).* 2021;2(3):100141. doi:10.1016/j.xinn.2021.100141.

- Kanehisa M, Sato Y, Kawashima M. KEGG mapping tools for uncovering hidden features in biological data. *Protein Sci.* 2022;31(1):47-53. doi:10.1002/pro.4172.
- Gillespie M, Jassal B, Stephan R, Milacic M, Rothfels K, Senff-Ribeiro A, et al. The Reactome pathway knowledgebase 2022. *Nucleic Acids Res.* 2022;50(D1):D687-D692. doi:10.1093/nar/gkab1028.
- Szklarczyk D, Kirsch R, Koutrouli M, Nastou K, Mehryary F, Hachilif R, et al. The STRING database in 2023: protein-protein association networks and functional enrichment analyses for any sequenced genome of interest. *Nucleic Acids Res.* 2023;51(D1):D638-D646. doi:10.1093/nar/gkac1000.
- Subramanian A, Tamayo P, Mootha VK, Mukherjee S, Ebert BL, Gillette MA, et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proc Natl Acad Sci U S A.* 2005;102(43):15545-15550. doi:10.1073/pnas.0506580102.
- Liberzon A, Birger C, Thorvaldsdottir H, Ghandi M, Mesirov JP, Tamayo P. The Molecular Signatures Database hallmark gene set collection. *Cell Syst.* 2015;1(6):417-425. doi:10.1016/j.cels.2015.12.004.
- Hong Y, Li J, Xu N, Shu W, Chen F, Mi Y, et al. Integrative genomic and single-cell framework identifies druggable targets for colorectal cancer precision therapy. *Front Immunol.* 2025;16:1604154. doi:10.3389/fimmu.2025.1604154.
- Hong Y, Wang Y, Wang D, Zhang Y, Zhang Y, Chen H, et al. Assessing male reproductive toxicity of environmental pollutant di-ethylhexyl phthalate with network toxicology and molecular docking strategy. *Reprod Toxicol.* 2024;130:108749. doi:10.1016/j.reprotox.2024.108749.
- Hong Y, Li J, Xu N, Wang Y, Shu W, Chen F, et al. Decoding per- and polyfluoroalkyl substances in hepatocellular carcinoma: a multi-omics and computational toxicology approach. *J Transl Med.* 2025;23:504. doi:10.1186/s12967-025-06517-z.

Hong Y, Chen H, Wang Y, Shu W, Xu N, Li J, et al. Integrative causal and single-cell analyses reveal
 510 genes linking endocrine-disrupting chemicals to male infertility. *Ecotoxicol Environ Saf.*
 2025;302:118709. doi:10.1016/j.ecoenv.2025.118709.

Hong Y, Wang Y, Shu W, Chen H, Li J, Xu N, et al. Prioritizing endocrine-disrupting chemicals
 targeting systemic lupus erythematosus through Mendelian randomization and colocalization.
Ecotoxicol Environ Saf. 2025;295:118126. doi:10.1016/j.ecoenv.2025.118126.

515 Lin Y, He Z, Ye J, Liu Z, She X, Gao X, et al. Progress in understanding the IL-6/STAT3 pathway in
 colorectal cancer. *Onco Targets Ther.* 2020;13:13055-13072. doi:10.2147/OTT.S278013.

Nagasaki T, Hara M, Nakanishi H, Takahashi H, Sato M, Takeyama H. Interleukin-6 released by colon
 cancer-associated fibroblasts is critical for tumour angiogenesis: anti-interleukin-6 receptor antibody
 suppressed angiogenesis and inhibited tumour-stroma interaction. *Br J Cancer.* 2014;110(2):469-478.
 520 doi:10.1038/bjc.2013.748.

Bromberg J, Wang TC. Inflammation and cancer: IL-6 and STAT3 complete the link. *Cancer Cell.*
 2009;15(2):79-80. doi:10.1016/j.ccr.2009.01.009.

Fisher DT, Appenheimer MM, Evans SS. The two faces of IL-6 in the tumor microenvironment.
Semin Immunol. 2014;26(1):38-47. doi:10.1016/j.smim.2014.01.008.

525 Soler MF, Merlotti A, Zlota A, Demaria S. New perspectives in cancer immunotherapy: targeting IL-6
 cytokine family. *J Immunother Cancer.* 2023;11(11):e007530. doi:10.1136/jitc-2023-007530.

Jassal B, Matthews L, Viteri G, Gong C, Lorente P, Fabregat A, et al. The Reactome pathway
 knowledgebase. *Nucleic Acids Res.* 2020;48(D1):D498-D503. doi:10.1093/nar/gkz1031.

Ren X, Chang C, Qi T, Yang P, Wang Y, Zhou X, Guan F, Li X. Clusterin Is a Prognostic Biomarker
 530 of Lower-Grade Gliomas and Is Associated with Immune Cell Infiltration. *Int J Mol Sci.*
 2023;24(17):13413. doi:10.3390/ijms241713413.

Xiao J, Zhang L, Su R, Zhao B, Dang Y, Zhao C, Wang S, Qi T, Ji F. MicroRNAs in breast cancer-
new frontiers in diagnosis, targeted therapy, and prognosis assessment. *Front Oncol.* 2025;15:1529907.
doi:10.3389/fonc.2025.1529907.

535 Figure legends

Figure 1. Study workflow and integrated evidence chain. The workflow summarizes the integrated
evidence-chain strategy used to evaluate IL6R in colorectal cancer, including pQTL-based Mendelian
randomization, IL6R-locus colocalization, TCGA COAD/READ transcriptomic and reconstructed
survival analyses, GSE146771 single-cell analysis, and support-layer network, enrichment and drug-
540 target analyses. The final conclusion emphasizes that MR and colocalization did not support strong
genetic causal or shared-variant evidence, whereas downstream transcriptomic and cellular analyses
linked IL6R to an immune-, myeloid- and inflammatory-associated tumour microenvironment.

Figure 2. Mendelian randomization analyses of genetically proxied circulating IL6R and colorectal
cancer risk. The figure summarizes method-specific MR estimates, statistical strength and
545 directionality across the primary FinnGen CRC endpoint and replication CRC GWAS endpoints using
readable endpoint labels. The estimates do not provide strong evidence that genetically proxied
circulating IL6R has a robust causal effect on colorectal cancer risk.

Figure 3. Colocalization analysis at the IL6R locus. Panel A shows posterior probabilities for coloc
hypotheses H0-H4. Panel B provides a compact posterior-evidence summary, and Panel C summarizes
550 the main colocalization conclusion. PP.H4 was approximately 0.048, indicating limited evidence that
circulating IL6R protein and colorectal cancer risk share a common causal variant at the IL6R locus.

Figure 4. TCGA expression and reconstructed OS/PFI survival analyses. The figure summarizes
TCGA patient-level IL6R expression distributions and reconstructed OS/PFI continuous Cox
estimates. These analyses support the interpretation that IL6R is not a robust standalone prognostic

555 marker in TCGA colorectal cancer despite its strong tumour-normal expression difference and
immune-microenvironment associations.

Figure 5. IL6R-associated immune infiltration patterns in TCGA colorectal cancer. The traceable
immune-infiltration figure summarizes correlations between IL6R expression and immune/stromal
infiltration estimates across deconvolution methods. Dot colour represents Spearman rho and dot size
560 represents statistical strength using a capped $-\log_{10}(\text{FDR})$ visual scale; full rho and FDR values are
retained in the source tables. The figure emphasizes myeloid, endothelial, neutrophil,
macrophage/monocyte and inflammatory tumour-microenvironment features.

Figure 6. Single-cell evidence for IL6-axis activity in GSE146771. The single-cell redraw summarizes
10x and Smart-seq2 cell-state structure, IL6-axis module evidence and CellChat-supported
565 communication evidence from traceable source tables. Smart-seq2 provides the clearer IL6-axis
communication signal, while the 10x dataset is retained as a sparsity-sensitive comparator. The figure
is interpreted as transcriptomic and cell-communication context rather than as independent causal
evidence.

Figure 7. IL6R-associated support-layer network, enrichment and drug-target evidence. The label-safe
570 figure combines an IL6R-centred support network with immune/inflammatory pathway enrichment
and exploratory drug-target support across IL6/IL6R/JAK/STAT and myeloid-associated axes. IL6R-
connected edges were retained in the final network rendering. Drug-target annotations are hypothesis-
generating and are not interpreted as evidence of clinically validated colorectal cancer therapy.

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